Pre-Analysis Plan
Doctor's Orders vs Divine Wisdom

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1 Introduction

How to deliver information most persuasively to individuals in developing countries is an important question with relevance not only for research, but also policy. Given the high rates of religiosity in many developing contexts, a recent literature considers the role of religious leaders and finds large impacts on behavior from information delivered by religious leaders. Yet, how religious leaders compare to secular leaders with more domain expertise, and whether utilizing religious leaders for information campaigns comes with unanticipated risks, remains understudied. Working with the Pakistan Ministry of Health (MOH), I compare the effectiveness of religious versus secular leaders in conducting health information campaigns in the context of Punjab, Pakistan. By experimentally varying who leads community meetings on the importance of handwashing as well as the framing of the message, I hope to answer the following research questions: 1) are religious leaders more effective than secular leaders at communicating information on health, 2) to what extent is the influence of religious and secular leaders domain-independent as opposed to domain-dependent, and 3) does receiving information from religious leaders have the unintended consequence of making them individuals' preferred choice for information in the future? Please refer to the AEA RCT Registry (AEARCTR-0018801) for full details of the interventions and study design.

2 Sample

I will sample individuals such that 40% are from urban areas and 60% are from rural areas, following the population distribution of Punjab. To limit my sample to regions where increasing rates of handwashing is most policy relevant, I will first select the three districts with the highest rates of under-5 diarrhea prevalence in the Punjab according to the 2017-18 Punjab Multiple Indicator Cluster Survey (MICS), subject to the constraint that no two

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districts belong to the same administrative division, which prevents the sample from being geographically concentrated in a single region of the province. Within each district, I will then randomly sample 10 communities - mouzas, or villages, in rural areas and census circles in urban areas - with probability proportional to population utilizing the 2023 Pakistan Population Census. I will stratify sampling such that 6 communities are from rural areas while 4 are urban. This will yield 30 communities in total, 18 rural and 12 urban, across 3 districts with high diarrhea prevalence in Punjab.

To identify couples within each community, I will drop 12 pins within the community's census boundaries, where each pin is a building drawn by simple random sampling without replacement from Google Open Buildings footprint dataset. This will ensure that pins land where people live rather than in unpopulated areas or fields and cause inclusion probabilities to be proportional to population/building density by construction. Per pin, enumerators will enroll the closest 5 couples, restricting eligibility to couples aged 18-50 and with at least one child below the age of 18. Both husband and wife will be asked to participate in the study, allowing me to evaluate heterogeneity in effects by gender. This procedure will give me 1800 couples in total, 60 selected from each community.

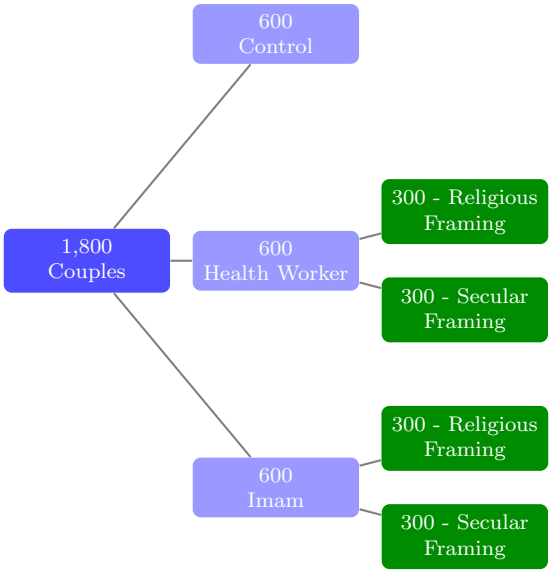
Once a couple has been enrolled in the study, they will be administered the baseline survey in person, separately for husband and wife. Once these baselines have been completed, couples will be randomized to treatment or control conditions, and treatments will be delivered. Treatments will be delivered in a staggered manner for logistical reasons and endline surveys will take place 1-4 weeks after the delivery of treatments.

3 Randomization

The unit of analysis is the individual, but randomization will be at the couple level. Couples within each community will be randomized to either a control condition, treatment 1 (in which the local imam delivers community meetings on handwashing), or treatment 2 (in which the local male health worker delivers community meetings on handwashing). Treatments will also be cross-randomized such that in half, the content is framed in fully religious language while in the other half it is in fully secular language. Randomization will be stratified by community so that in each community, every couple has an equal chance of being assigned to any treatment group.

The randomization is shown visually in Figure 1 below.

Figure 1: Randomization



The treatments will consist of two 15-20 minute community meetings conducted over the course of two weeks in which the assigned leader memorizes and then delivers scripts on the importance of handwashing. Each meeting will cover distinct content on the importance of handwashing, rather than repeating the same content. The meetings will be recorded and transcribed, and their content quantified to evaluate the extent of compliance for each type of leader.

4 Outcomes

There will be two broad sets of outcomes: 1) behavioral outcomes, 2) self-reported behavior, beliefs, and knowledge. The behavioral outcomes will be measured only at endline, while the self-reports will be measured at both baseline and endline.

4.1 Primary Outcomes

My primary outcomes are designed to capture changes in actual hand hygiene behavior and changes in who individuals prefer for information following treatments.

To evaluate effects on actual hand hygiene, respondents will be administered a single ATP test at endline. The test relies on collecting Adenosine Triphosphate (ATP) from the surfaces of hands which can then be numerically measured using a precise device called a Luminometer. To allow for ease of interpretability, the ATP reading will be standardized using the control group mean and standard deviation and shown with sign reversed so that higher values denote cleaner hands. I will also evaluate effects on log ATP readings for robustness.

To evaluate whether individuals update who they prefer for information, at endline they will be offered information in the form of videos on a secular topic that is the health worker's expertise (family planning) and a religious topic that is the imam's expertise (religious giving). The imam and health worker will be free to say what they like in each 5-10 minute video and the respondent will have to choose whose video they would like to see for each domain. Individuals' choices will then serve to proxy whether the treatments change who individuals prefer for religious and secular information. The two choices yield two binary outcomes: an indicator for choosing the imam over the health worker on the secular topic, and an indicator for choosing the imam over the health worker on the religious topic.

Both outcomes will be collected in person only at endline. The ATP test will be administered in an unannounced visit: an enumerator will arrive at the household without prior notice so that the hand hygiene captured is authentic rather than prepared in anticipation of the visit. The choice of videos will be offered at the end of the full endline survey.

4.2 Secondary Outcomes

In addition to these primary outcomes, I also hope to collect data on secondary outcomes that can further help test my hypotheses and shed light on mechanisms. All indices will be constructed following Anderson (2008).

- Self-reported knowledge: To test knowledge, the treatments will contain factual information which will be tested at both baseline and endline through four questions that test respondents' secular and religious knowledge about handwashing. The knowledge outcome will be an index summarizing closeness to the actual answer for all four questions. Respondents also report, on a 0-10 scale, how confident they are that their four answers are accurate.

- Self-reported health beliefs and behavior: To capture subtle changes in beliefs and behavior, respondents will be asked to self-report their health beliefs and behavior. These will be summarized in two indices: a health behavior index (e.g. self-reported handwashing frequency, soap use, handwashing behavior of children etc.) and a health beliefs index (e.g. whether handwashing is more important than a balanced diet for staying healthy). Adjacent health behaviors such as toothbrushing will form a third, separate index to test whether persuasion spills over to health behaviors not targeted by the treatments. Respondents will also report the number of days in the past week on which they and their youngest child suffered from diarrhea or felt sick.

- Second-order beliefs about social norms: I will ask respondents to report their perceptions of others' beliefs and behaviors with respect to handwashing. This will allow me to test whether any treatment effects operate through perceived social norms.

- Unintended negative consequences: I will explore additional unintended negative consequences of the religious leader treatments through two vignettes: 1) the first will

compare trust in state versus religious institutions by presenting a hypothetical person who is deciding between state and religious options for a range of services and asking respondents to help him choose between the two, 2) the second will evaluate beliefs about the efficacy of handwashing relative to religiosity for preventing illness, as opposed to fatalism, by asking respondents to predict the number of children who will suffer from diarrhea across men who differ in the handwashing behavior of their children and in their religiosity and also to predict the number of children that will fall sick despite all preventive behaviors due to divine fate.

5 Analysis

5.1 Main Specification

My main specification that captures the pooled impact of the interventions is:

$$y_{ick} = \alpha + \beta_1 \text{Imam}_c + \beta_2 \text{HW}_c + \theta_k + \epsilon_{ick} \tag{1}$$

where y_{ick} is the outcome of interest for individual i of couple c in community k measured at endline, Imam_c and HW_c are indicators for whether the couple was assigned to the imam meetings or health worker meetings treatment respectively, θ_k represents community fixed effects, and ϵ_{ick} is the error term. All standard errors will be clustered at the couple level, the unit of randomization. I will also test the null hypothesis $\beta_1 = \beta_2$ to directly compare the effectiveness of religious versus secular leaders. I will first run this specification pooling husbands and wives and then disaggregate to explore heterogeneous effects by gender.

I will also run the same specification but with controls:

$$y_{ick} = \alpha + \beta_1 \text{Imam}_c + \beta_2 \text{HW}_c + X_{ick}\gamma + \theta_k + \epsilon_{ick} \tag{2}$$

where X_{ick} is a set of control variables which I will select using LASSO and baseline data. For outcomes available at baseline, X_{ick} will include the baseline value of the outcome.

In addition to these main specifications, to evaluate impacts of religious versus secular framing, I also hope to disaggregate my treatment indicators as follows:

$$y_{ick} = \alpha + \beta_1 \text{ImamRel}_c + \beta_2 \text{ImamSec}_c + \beta_3 \text{HWRel}_c + \beta_4 \text{HWSec}_c + X_{ick}\gamma + \theta_k + \epsilon_{ick} \tag{3}$$

where ImamRel_c, ImamSec_c, HWRel_c, and HWSec_c are indicators for the four cross-randomized treatment cells. By comparing whether β_1 and β_2 as well as β_3 and β_4 are significantly different from each other, I hope to test whether religious versus secular framing produces statistically distinguishable effects for imams relative to health workers.

5.2 Treatment Effect Heterogeneity

Following my theoretical model, the extent of updating depends on the magnitude of the difference between the signal and prior means, religiosity, and the precision of the signal.

Accordingly, I hope to evaluate treatment effect heterogeneity by respondents': 1) baseline priors about handwashing, 2) baseline religiosity, 3) baseline trust in the imam relative to the health worker.

I will also explore heterogeneity by the rural/urban status of the community, district, basic demographics (e.g. age, gender, income), characteristics of the leader delivering the treatment, as well as baseline health beliefs and behavior. I will also explore persistence by looking at decay in treatment effects with time since treatment. Finally, I will test for spillovers by comparing outcomes for control couples residing close to treatment couples versus those further away.